C.W

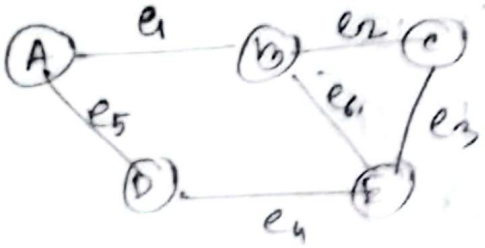
05.10.2024

Graph Representation: (6 marks)

→ incedence Matrix

→ Adjecency Matrix

Ex:



Incedence Matrix:

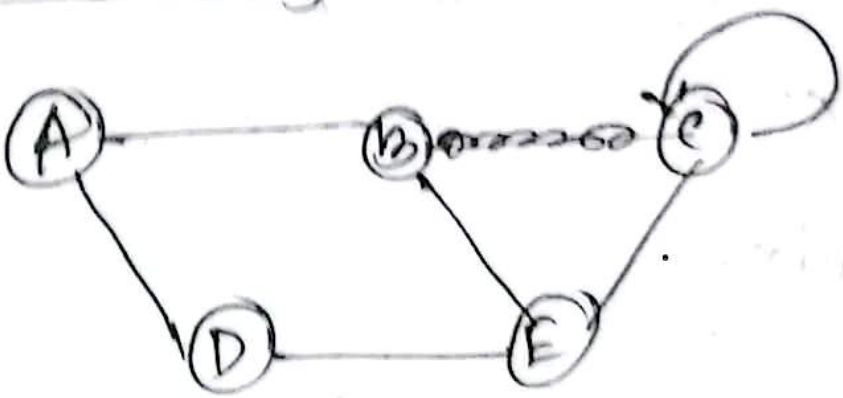
	e_1	e_2	e_3	e_4	e_5	e_6	e_7
A	1	0	0	0	1	0	
B	1	1	0	0	0	1	
C	0	1	1	0	0	0	
D	0	0	0	1	1	0	
E	0	0	1	1	0	1	

∴ Matrix:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$$

CamScanner

⊛ Adjecency Matrix:



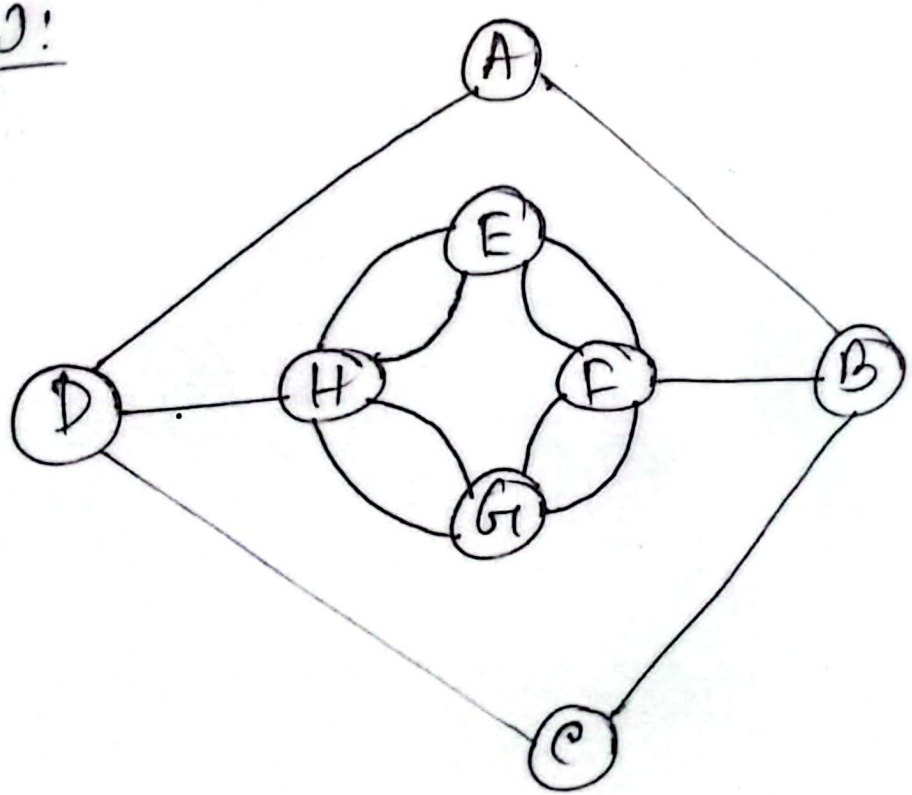
	A	B	C	D	E
A	0	1	0	1	0
B	1	0	0	0	1
C	0	0	1	0	1
D	1	0	0	0	1
E	0	1	1	1	0

Matrix:

$$\begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

CamScanner

HW:



; find two types of matrixes of the graph.